Note

For practical purposes, the sample may be regarded as large if $n > 30$.

8.3.2 Testing Procedure : Large sample theory and test of significants for single mean

The following are the steps involved in hypothesis testing problems.

1. Null hypothesis: Set up the null hypothesis H_0
2. Alternative hypothesis: Set up the alternative hypothesis . This will enable us to decide whether we have to use two tailed test or single tailed test (right or left tailed)
3. Level of significance: Choose the appropriate level of significant (α) depending on the reliability of the estimates and permissible risk. This is to be fixed before sample is drawn. i.e., α is fixed in advance.
4. Test statistic : Compute the test statistic using

$$Z = \frac{t - E(t)}{\sqrt{Var(t)}} = \frac{t - E(t)}{S.E.(t)} \sim N(0,1) \text{ as } n \to \infty$$

5. Conclusion: We compare the computed value of Z in step 4 with the significant value or critical value or table value Z_α at the given level of significance .

   (i) If $|Z| < Z_\alpha$ i.e., if the calculated value of is less than critical value we say it is not significant. This may due to fluctuations of sampling and sample data do not provide us sufficient evidence against the null hypothesis which may therefore be accepted.

   (ii) If $|Z| > Z_\alpha$ i.e., if the calculated value of is greater than critical value Z_α then we say it is significant and the null hypothesis is rejected at level of significance α .

Test of significance for single mean

Let $x_i, (i = 1,2,3,...,n)$ is a random sample of size from a normal population with mean μ and variance σ^2 then the sample mean is distributed normally with mean μ and variance $\frac{\sigma^2}{n}$, i.e., $\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$. Thus for large samples, the standard normal variate corresponding to $\bar{x}$ is :

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Under the null hypothesis that the sample has been drawn from a population with mean and variance σ^2, i.e., there is no significant difference between the sample mean $(\bar{x})$ and the population mean (μ), the test statistic (for large samples) is:

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Remark:

If the population standard deviation σ is unknown then we use its estimate provided by the sample variance given by $\hat{\sigma}^2 = s_2 \Rightarrow \hat{\sigma} = s$.

Example 8.14

An auto company decided to introduce a new six cylinder car whose mean petrol consumption is claimed to be lower than that of the existing auto engine. It was found that the mean petrol consumption for the 50 cars was 10 km per litre with a standard deviation of 3.5 km per litre. Test at 5% level of significance, whether the claim of the new car petrol consumption is 9.5 km per litre on the average is acceptable.

Solution:

Sample size n =50 Sample mean $\bar{x}$ = 10 km Sample standard deviation s = 3.5 km

Population mean μ =9.5 km

Since population SD is unknown we consider σ = s

The sample is a large sample and so we apply Z-test.

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